

**4.0.1 ISRO CSE 2020 | Question: 60**

A given grammar is called ambiguous if

- A. two or more productions have the same non-terminal on the left hand side
- B. a derivation tree has more than one associated sentence
- C. there is a sentence with more than one derivation tree corresponding to it
- D. brackets are not present in the grammar

isro-2020 compiler-design ambiguous-grammar easy

Answer key

4.0.2 ISRO CSE 2023 | Question: 21

Which sequence below best describes the order in which the phases of a compiler typically work?

- A. Syntax analysis, lexical analysis, χ semantic analysis, code optimization, intermediate code generation
- B. Lexical analysis, syntax analysis, semantic analysis, code optimization, intermediate code generation
- C. Syntax analysis, lexical analysis, syntax analysis, semantic analysis, syntax analysis, intermediate code generation, code optimization
- D. Lexical analysis, syntax analysis, semantic analysis, intermediate code generation, code optimization

isro-cse-2023 compiler-design

Answer key

4.1**Assembler (2)****4.1.1 Assembler: ISRO CSE 2016 | Question: 43**

A simple two-pass assembler does which of the following in the first pass:

- A. Checks to see if the instructions are legal in the current assembly mode
- B. It allocates space for the literals.
- C. It builds the symbol table for the symbols and their values.
- D. All of these

compiler-design assembler isro2016

Answer key

4.1.2 Assembler: ISRO CSE 2020 | Question: 14

In a two-pass assembler, resolution of subroutine calls and inclusion of labels in the symbol table is done during

- A. second pass
- B. first pass and second pass respectively
- C. second pass and first pass respectively
- D. first pass

Answer key **4.2 Code Optimization (9)****4.2.1 Code Optimization: ISRO CSE 2008 | Question: 68**

Which of the following class of statement usually produces no executable code when compiled?

- | | |
|--------------------------------|--------------------------|
| A. declaration | B. assignment statements |
| C. input and output statements | D. structural statements |

isro2008 compiler-design code-optimization

Answer key **4.2.2 Code Optimization: ISRO CSE 2009 | Question: 43**

Substitution of values for names (whose values are constants) is done in

- | | |
|-----------------------|-----------------------|
| A. Local optimization | B. Loop optimization |
| C. Constant folding | D. Strength reduction |

isro2009 compiler-design code-optimization

Answer key **4.2.3 Code Optimization: ISRO CSE 2011 | Question: 13**

Which of the following statements about peephole optimization is False?

- A. It is applied to a small part of the code
- B. It can be used to optimize intermediate code
- C. To get the best out of this, it has to be applied repeatedly
- D. It can be applied to the portion of the code that is not contiguous

isro2011 compiler-design code-optimization

Answer key **4.2.4 Code Optimization: ISRO CSE 2011 | Question: 3**

In compiler terminology reduction in strength means

- A. Replacing run time computation by compile time computation
- B. Removing loop invariant computation
- C. Removing common subexpressions
- D. replacing a costly operation by a relatively cheaper one

isro2011 compiler-design code-optimization

Answer key **4.2.5 Code Optimization: ISRO CSE 2016 | Question: 44**

Peephole optimization is form of

- | | |
|----------------------|-----------------------|
| A. Loop optimization | B. Local optimization |
| C. Constant folding | D. Data flow analysis |

compiler-design code-optimization isro2016

Answer key 

4.2.6 Code Optimization: ISRO CSE 2018 | Question: 29



DU-chains(Definition-Use) in compiler design:

- a. consist of a definition of a variable and all its uses, reachable from that definition
- b. are created using a form static code analysis
- c. are prerequisite for many compiler optimization including constant propagation and common sub-expression elimination
- d. All of the above

isro2018 compiler-design code-optimization

Answer key

4.2.7 Code Optimization: ISRO CSE 2018 | Question: 30



Which of the following comment about peep-hole optimization is true?

- a. It is applied to small part of the code and applied repeatedly
- b. It can be used to optimize intermediate code
- c. It can be applied to a portion of the code that is not contiguous
- d. It is applied in symbol table to optimize the memory requirements.

isro2018 compiler-design code-optimization

Answer key

4.2.8 Code Optimization: ISRO CSE 2020 | Question: 42



Which of the following is a type of a out-of-order execution, with the reordering done by a compiler

- A. loop unrolling
- B. dead code elimination
- C. strength reduction
- D. software pipelining

isro-2020 compiler-design code-optimization normal

Answer key

4.2.9 Code Optimization: ISRO-DEC2017-67



Consider the code segment

```
int i,j,x,y,m,n;
n = 20;
for(i=0;i<n;i++)
{
    for(j=0;j<n;j++)
    {
        if(i%2)
        {
            x += ((4*j)+5*i);
            y += (7+4*j);
        }
    }
}
m=x+y;
```

Which one of the following is false?

- A. The code contains loop invariant computation
- B. There is scope of common sub-expression elimination in this code
- C. There is scope of strength reduction in this code

D. There is scope of dead code elimination in this code

isrodec2017 code-optimization compiler-design

Answer key

4.3 Compilation Phases (1)

4.3.1 Compilation Phases: ISRO CSE 2020 | Question: 17



Which one indicates a technique of building cross compilers?

- A. Beta cross B. Canadian cross C. Mexican cross D. X-cross

isro-2020 compiler-design compilation-phases normal

Answer key

4.4 Compiler tokenization (1)

4.4.1 Compiler tokenization: ISRO CSE 2020 | Question: 13



The number of tokens in the following C code segment is

```
switch(inputvalue)
{
    case 1 : b =c*d; break;
    default : b =b++; break;
}
```

- A. 27 B. 29 C. 26 D. 24

isro-2020 compiler-design compiler-tokenization easy

Answer key

4.5 Context Free Grammar (1)

4.5.1 Context Free Grammar: ISRO CSE 2018 | Question: 64



A particular BNF definition for a "word is given by the following rules.

```
<word> ::= <letter> | <letter> <charpair> | <letter> <intpair>
<charpair> ::= <letter> <letter> | <charpair> <letter> <letter>
<intpair> ::= <integer> <integer> | <intpair> <integer> <integer>
<letter> ::= a | b | c | ..... | Y | Z
<integer> ::= 0 | 1 | 2 | ..... | 9
```

Which of the following lexical entries can be derived from < word >?

- I. pick
II. picks
III. c44

- A. I, II and III B. I and II only C. I and III only D. II and III only

isro2018 compiler-design grammar context-free-grammar

Answer key

4.6 Expression Evaluation (1)

4.6.1 Expression Evaluation: ISRO CSE 2009 | Question: 32



The expression $1 * 2 \wedge 3 * 4 \wedge 5 * 6$ will be evaluated as

- A. 32^{30}
- B. 162^{30}
- C. 49152
- D. 173458

In original question 5 is not given.

isro2009 compiler-design expression-evaluation

Answer key

4.7

Grammar (5)

4.7.1 Grammar: GATE CSE 2006 | Question: 32, ISRO2016-35



Consider the following statements about the context free grammar

$$G = \{S \rightarrow SS, S \rightarrow ab, S \rightarrow ba, S \rightarrow \epsilon\}$$

- I. G is ambiguous
- II. G produces all strings with equal number of a 's and b 's
- III. G can be accepted by a deterministic PDA.

Which combination below expresses all the true statements about G ?

- A. I only
- B. I and III only
- C. II and III only
- D. I, II and III

gatecse-2006 compiler-design grammar normal isro2016

Answer key

4.7.2 Grammar: ISRO CSE 2014 | Question: 52



Consider the following grammar.

- $S \rightarrow AB$
- $A \rightarrow a$
- $A \rightarrow BaB$
- $B \rightarrow bbA$

Which of the following statements is FALSE?

- A. The length of every string produced by this grammar is even
- B. No string produced by this grammar has three consecutive a 's
- C. The length of substring produced by B is always odd
- D. No string produced by this grammar has four consecutive b 's

isro2014 compiler-design grammar parsing

Answer key

4.7.3 Grammar: ISRO CSE 2020 | Question: 35



Given the grammar

- $s \rightarrow T^*S \mid T$
- $T \rightarrow U + T \mid U$
- $U \rightarrow a \mid b$

Which of the following statements is wrong?

- A. Grammar is not ambiguous
- B. Priority of $+$ over $*$ is ensured
- C. Right to left evaluation of $*$ and $+$ happens
- D. None of these

isro-2020 compiler-design grammar easy

Answer key

4.7.4 Grammar: ISRO CSE 2020 | Question: 75



A grammar is defined as

- $A \rightarrow BC$
- $B \rightarrow x \mid Bx$
- $C \rightarrow B \mid D$
- $D \rightarrow y \mid Ey$
- $E \rightarrow z$

The non terminal alphabet of the grammar is

- A. $\{A, B, C, D, E\}$
- B. $\{B, C, D, E\}$
- C. $\{A, B, C, D, E, x, y, z\}$
- D. $\{x, y, z\}$

isro-2020 compiler-design grammar parsing easy

Answer key

4.7.5 Grammar: ISRO-2013-6



Which of the following productions eliminate left recursion in the productions given below:

$$S \rightarrow Aa \mid b$$

$$A \rightarrow Ac \mid Sd \mid \epsilon$$

- A. $S \rightarrow Aa \mid b, A \rightarrow bdA', A' \rightarrow A'c \mid A'ba \mid A \mid \epsilon$
- B. $S \rightarrow Aa \mid b, A \rightarrow A' \mid bdA', A' \rightarrow cA' \mid adA' \mid \epsilon$
- C. $S \rightarrow Aa \mid b, A \rightarrow A'c \mid A'd, A' \rightarrow bdA' \mid cA \mid \epsilon$
- D. $S \rightarrow Aa \mid b, A \rightarrow cA' \mid adA' \mid bdA', A' \rightarrow A \mid \epsilon$

isro2013 parsing grammar

Answer key

4.8

Heap Allocation (1)

4.8.1 Heap Allocation: ISRO-DEC2017-66



Which languages necessarily need heap allocation in the runtime environment?

- A. Those that support recursion
- B. Those that use dynamic scoping
- C. Those that use global variables
- D. Those that allow dynamic data structures

isrodec2017 runtime-environment heap-allocation operating-system data-structures

Answer key 

4.9

Intermediate Code (2)

4.9.1 Intermediate Code: ISRO CSE 2008 | Question: 38



Relative to the program translated by a compiler, the same program when interpreted runs

- A. Faster
- B. Slower
- C. At the same speed
- D. May be faster or slower

isro2008 compiler-design intermediate-code

Answer key 

4.9.2 Intermediate Code: ISRO CSE 2018 | Question: 28



Incremental-Compiler is a compiler:

- a. which is written in a language that is different from the source language
- b. compiles the whole source code to generate object code afresh
- c. compiles only those portion of source code that have been modified
- d. that runs on one machine but produces object code for another machine

isro2018 compiler-design intermediate-code

Answer key 

4.10

Lexical Analysis (2)

4.10.1 Lexical Analysis: GATE CSE 2000 | Question: 1.18, ISRO2015-25



The number of tokens in the following C statement is

```
printf("i=%d, &i=%x", i, &i);
```

- A. 3
- B. 26
- C. 10
- D. 21

gatecse-2000 compiler-design lexical-analysis easy isro2015

Answer key 

4.10.2 Lexical Analysis: ISRO 2017-79



The output of a lexical analyzer is

- A. A parse tree
- B. Intermediate code
- C. Machine code
- D. A stream of tokens

isro2017 compiler-design lexical-analysis

Answer key 

4.11

Memory Management (1)

4.11.1 Memory Management: ISRO CSE 2023 | Question: 31



As part of segmentation memory allocation strategy, which of the following assigns segment numbers for various segments of a program like code segment and data segment?

- A. Compiler
- B. Assembler
- C. Loader
- D. Linker

Answer key

4.12

Operator Grammar (1)

4.12.1 Operator Grammar: ISRO2015-26



Which grammar rules violate the requirement of the operator grammar? A, B, C are variables and a, b, c are terminals

- i. $A \rightarrow BC$
- ii. $A \rightarrow CcBb$
- iii. $A \rightarrow BaC$
- iv. $A \rightarrow \epsilon$

- a. (i) only b. (i) and (ii) c. (i) and (iii) d. (i) and (iv)

isro2015 compiler-design operator-grammar parsing

Answer key

4.13

Operator Precedence (1)

4.13.1 Operator Precedence: GATE CSE 2000 | Question: 2.21, ISRO2015-24



Given the following expression grammar:

$$E \rightarrow E * F \mid F + E \mid F$$

$$F \rightarrow F - F \mid id$$

Which of the following is true?

- A. $*$ has higher precedence than $+$ B. $-$ has higher precedence than $*$
 C. $+$ and $-$ have same precedence D. $+$ has higher precedence than $*$

gatecse-2000 operator-precedence normal compiler-design isro2015

Answer key

4.14

Parameter Passing (2)

4.14.1 Parameter Passing: GATE CSE 2004 | Question: 2, ISRO2017-54



Consider the following function

```
void swap(int a, int b)
{
    int temp;
    temp = a;
    a = b;
    b = temp;
}
```

In order to exchange the values of two variables x and y .

- A. call $swap(x, y)$
 B. call $swap(\&x, \&y)$
 C. $swap(x, y)$ cannot be used as it does not return any value

D. $swap(x, y)$ cannot be used as the parameters are passed by value

gatecse-2004 compiler-design programming-in-c parameter-passing easy isro2017 runtime-environment

Answer key

4.14.2 Parameter Passing: ISRO2015-80



Consider the following code fragment

```
void foo(int x, int y)
{
    x+=y;
    y+=x;
}
main()
{
    int x=5.5;
    foo(x,x);
}
```

What is the final value of x in both call by value and call by reference, respectively?

- A. 5 and 16 B. 5 and 12 C. 5 and 20 D. 12 and 20

isro2015 compiler-design runtime-environment parameter-passing

Answer key

4.15 Parsing (9)

4.15.1 Parsing: ISRO CSE 2011 | Question: 43



Which variable does not drive a terminal string in grammar?

- $S \rightarrow AB$
- $A \rightarrow a$
- $B \rightarrow b$
- $B \rightarrow C$

- A. A B. B C. C D. S

isro2011 compiler-design parsing

Answer key

4.15.2 Parsing: ISRO CSE 2014 | Question: 5



What is the number of steps required to derive the string $((() ()) ())$ for the following grammar?

- $S \rightarrow SS$
- $S \rightarrow (S)$
- $S \rightarrow \epsilon$

- A. 10 B. 12
C. 15 D. 16

isro2014 compiler-design parsing

Answer key

4.15.3 Parsing: ISRO CSE 2016 | Question: 40



Recursive descent parsing is an example of

- A. Top-down parsers
- B. Bottom-up parsers
- C. Predictive parsers
- D. None of the above

isro2016 compiler-design parsing

Answer key

4.15.4 Parsing: ISRO CSE 2016 | Question: 41



A top-down parser generates

- A. Rightmost Derivation
- B. Rightmost derivation in reverse
- C. Leftmost derivation
- D. Leftmost derivation in reverse

isro2016 compiler-design parsing

Answer key

4.15.5 Parsing: ISRO CSE 2023 | Question: 22



In the context of parsing techniques, LL(1) refers to:

- A. Grammars that parse the input from left to right, produce a rightmost derivation, with a lookahead of 1 symbol.
- B. Grammars that parse the input from left to right, produce a leftmost derivation, with a lookahead of 1 symbol.
- C. Grammars that parse the input from left to right, and backtrack at most once.
- D. Grammars that parse the input from right to left, produce a leftmost derivation, and backtrack at most once.

isro-cse-2023 compiler-design parsing theory-of-computation

Answer key

4.15.6 Parsing: ISRO-2013-5



Shift reduce parsing belongs to a class of

- A. Bottom up parsing.
- B. Top down parsing.
- C. Recursive parsing.
- D. Predictive parsing.

isro2013 parsing

Answer key

4.15.7 Parsing: ISRO2015-27



Which one of the following is a top-down parser?

- A. Recursive descent parser
- B. Shift left associative parser
- C. $SLR(k)$ parser
- D. $LR(k)$ parser

isro2015 compiler-design parsing

Answer key

4.15.8 Parsing: ISRO2015-28



Yacc stands for

- A. yet accept compiler constructs
- B. yet accept compiler compiler
- C. yet another compiler constructs
- D. yet another compiler compiler

Answer key

4.15.9 Parsing: ISRO2015-29



Which statement is true?

- A. LALR parser is more powerful and costly as compare to other parsers
- B. All CFG's are LP and not all grammars are uniquely defined
- C. Every SLR grammar is unambiguous but not every unambiguous grammar is SLR
- D. LR(K) is the most general back tracking shift reduce parsing method

Answer key

4.16 Runtime Environment (1)

4.16.1 Runtime Environment: GATE CSE 1998 | Question: 1.25, ISRO2008-41



In a resident – OS computer, which of the following systems must reside in the main memory under all situations?

- A. Assembler
- B. Linker
- C. Loader
- D. Compiler

Answer key

4.17 Symbol Table (1)

4.17.1 Symbol Table: ISRO CSE 2011 | Question: 78



A symbol table of length 152 is processing 25 entries at any instant. What is occupation density?

- A. 0.164
- B. 127
- C. 8.06
- D. 6.08

Answer key

4.18 Variable Binding (1)

4.18.1 Variable Binding: ISRO-2013-7



Consider the following psuedocode:

```
x: integer := 1
y: integer := 2
procedure add
  x := x + y
procedure second (P: Procedure)
  x: integer := 2
  p()
procedure first
  y: integer := 3
  second (add)
first ()
write_integer(x)
```

What does it print if the language uses dynamic scoping with deep binding?

- A. 2
- B. 3
- C. 4
- D. 5

Answer key

Answer Keys

4.0.1	C
4.2.2	C
4.2.7	X
4.5.1	D
4.7.4	A
4.10.1	C
4.14.1	D
4.15.4	C
4.15.9	C

4.0.2	D
4.2.3	X
4.2.8	D
4.6.1	C
4.7.5	B
4.10.2	D
4.14.2	C
4.15.5	B
4.16.1	C

4.1.1	D
4.2.4	D
4.2.9	D
4.7.1	B
4.8.1	D
4.11.1	C
4.15.1	C
4.15.6	A
4.17.1	A

4.1.2	C
4.2.5	B
4.3.1	B
4.7.2	D
4.9.1	B
4.12.1	D
4.15.2	A
4.15.7	A
4.18.1	C

4.2.1	A
4.2.6	D
4.4.1	C
4.7.3	D
4.9.2	C
4.13.1	B
4.15.3	A
4.15.8	D